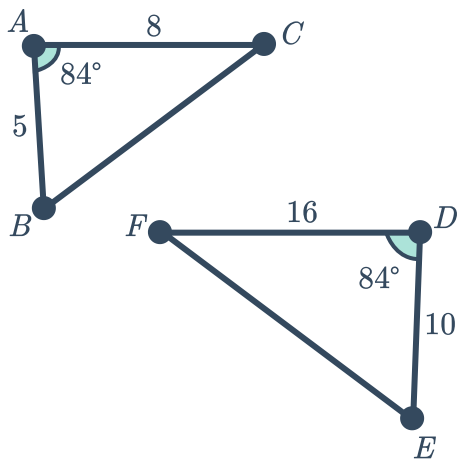
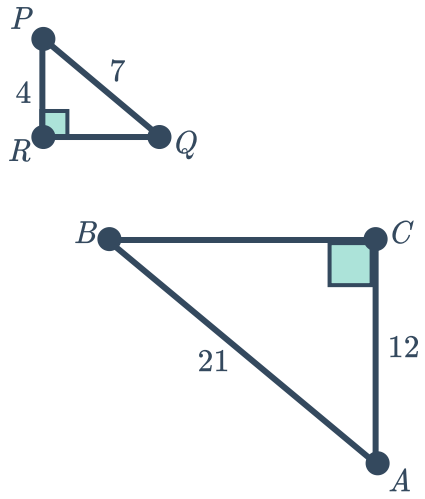
# Similar triangles

**1** State whether the following pairs of triangles are similar, and if so, state the similarity test that proves this.

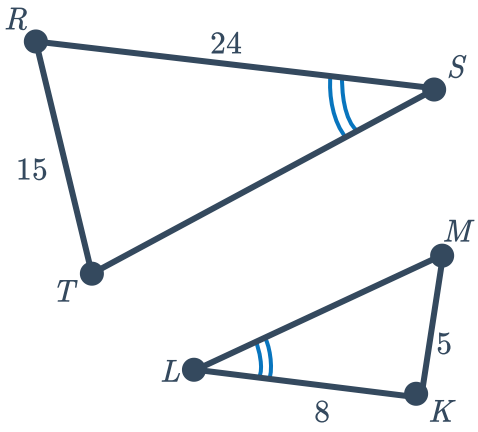
**a**



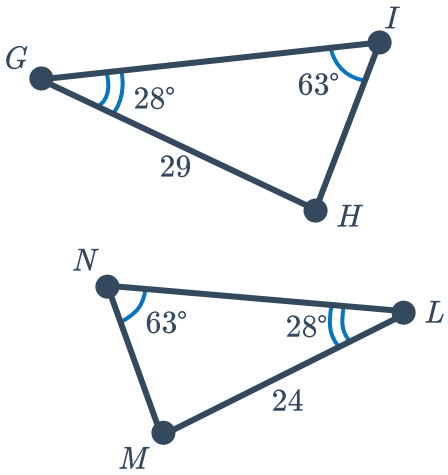

**b**



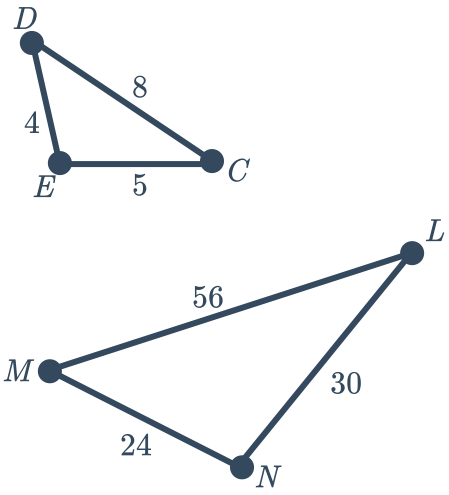

**c**



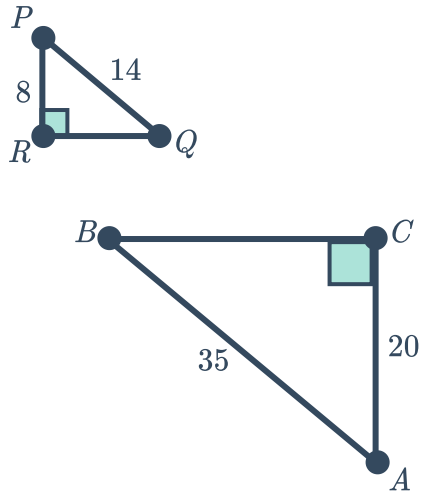

**d**




**e**

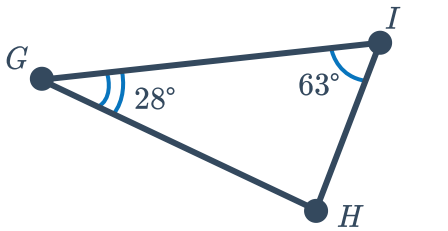


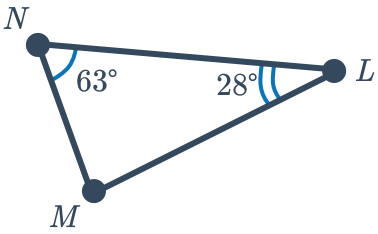

**f**



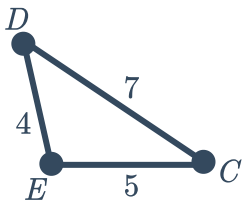

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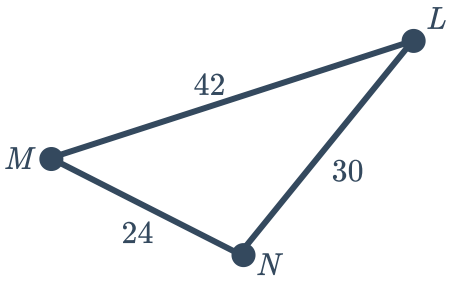
g




h



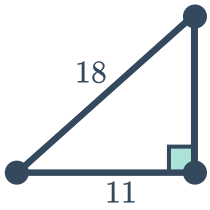

2 For each of the following sets of triangles:

i Identify which two are similar.

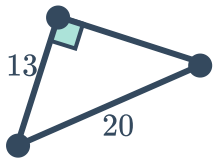
ii Identify the similarity test that the pair satisfy: AAA, SSS, SAS, or RHS.

a

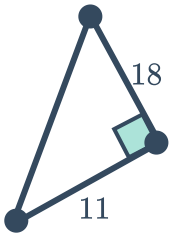
A



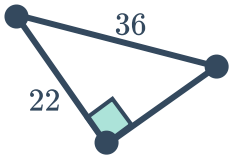

B




C




D

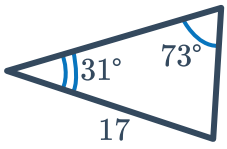



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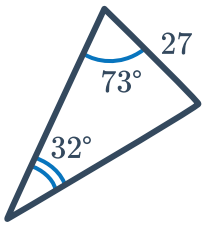
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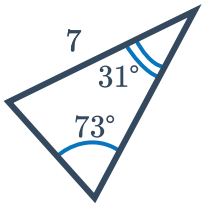

B




C

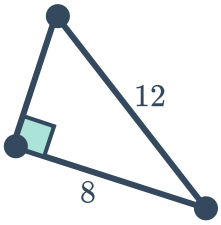



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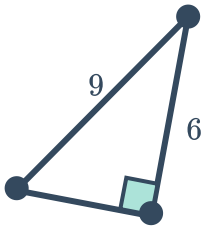



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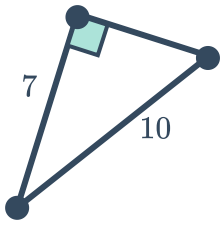
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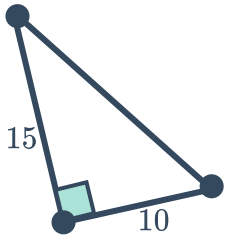

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C



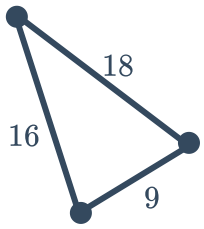

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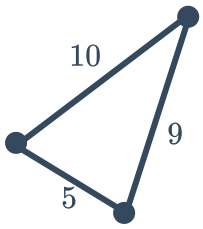

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d

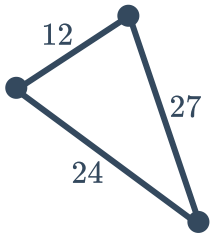
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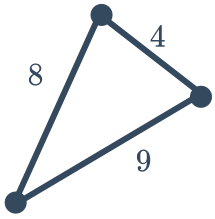

B




C

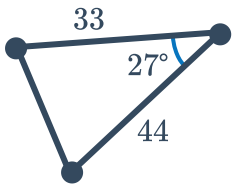



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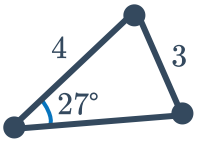



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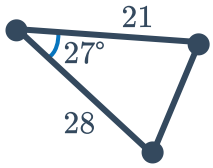
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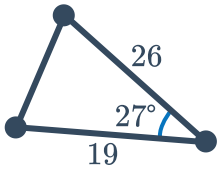

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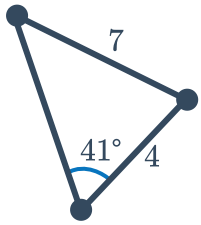



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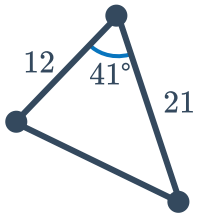



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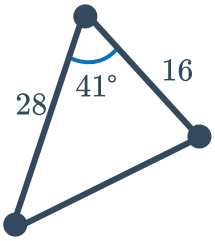
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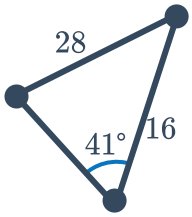

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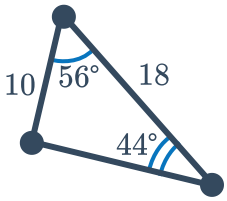



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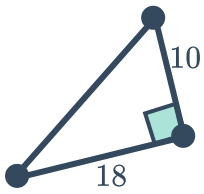



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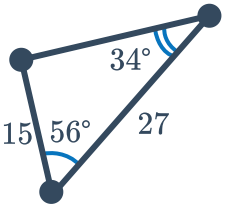
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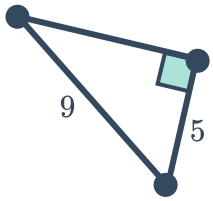

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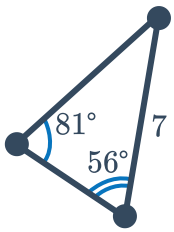



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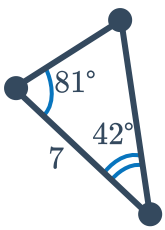



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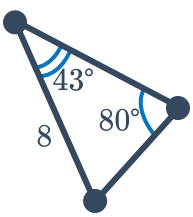
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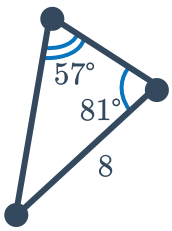

B




C

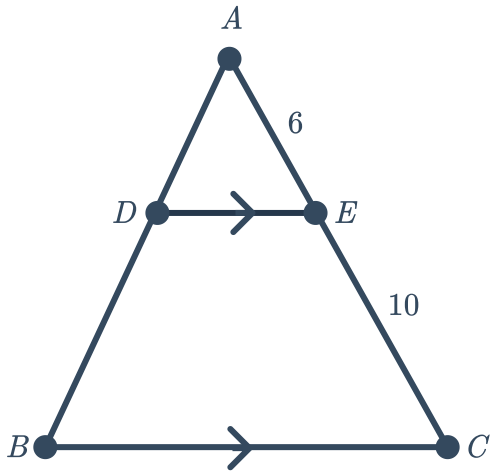



D




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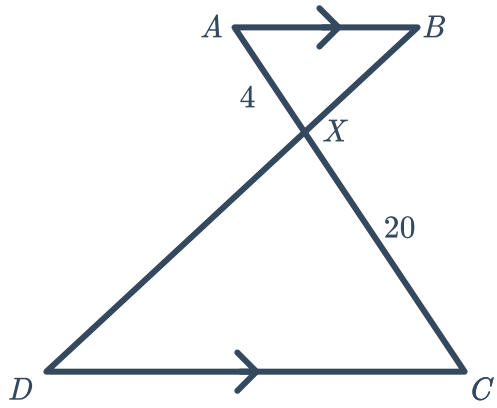
**3** Consider the following diagram:




**a** Identify two similar triangles and state the similarity test that they satisfy.

**b** Given that $DE = 7$, what is the length of $BC$?

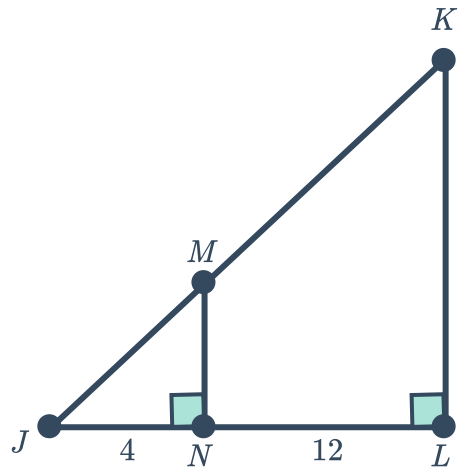
**4** Consider the following diagram:




**a** State the similarity test that triangles $\triangle ABX$ and $\triangle CDX$ satisfy.

**b** Given that $AB = 5$, what is the length of $CD$?

**5** Consider the following diagram:



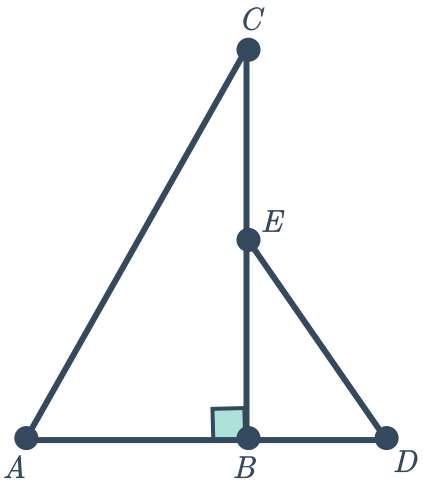

**a** Identify two similar triangles and state the similarity test that they satisfy.

**b** Given that $KL = 20$, what is the length of $MN$?

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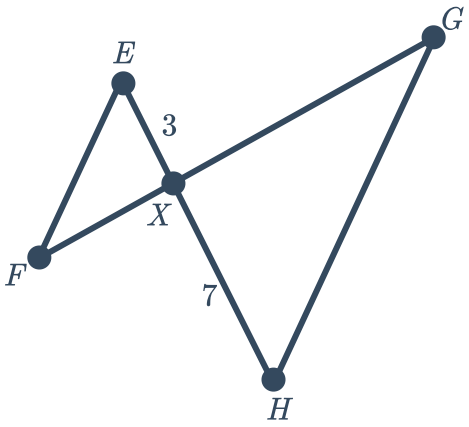
**6** Consider the following diagram, where $AD$ is a straight line:

What information is required to know that triangles $\triangle ABC$ and $\triangle DBE$ are in fact similar?




**7** Consider the following diagram, where $EH$ and $FG$ are straight lines:

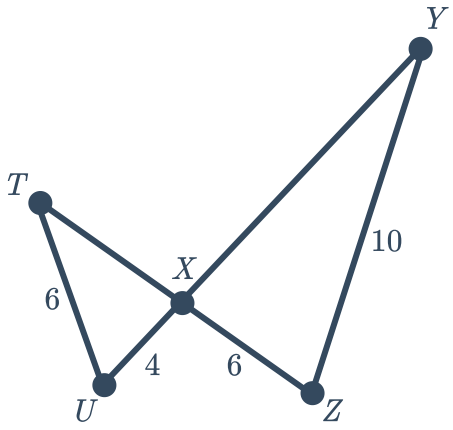
What information is required to know that triangles $\triangle EFX$ and $\triangle GHX$ are in fact similar?




**8** Consider the following diagram, where $TZ$ and $UY$ are non-parallel straight lines:

**a** Explain why the triangles $\triangle TUX$ and $\triangle YZX$ are not similar.

**b** What length does $YZ$ need to be for the two triangles to be similar?



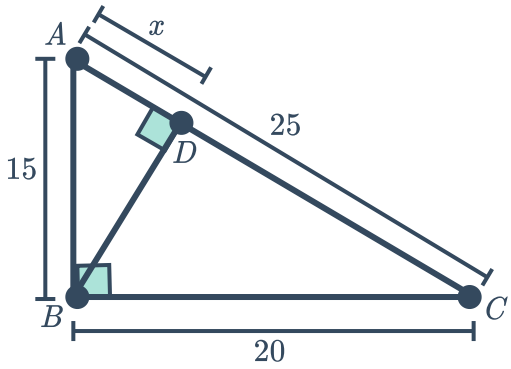

**9** Peter constructs a ramp in the shape of a right-angled triangle. To give the ramp more strength, he wants to place a brace from the base point across to the hypotenuse. The brace should meet the hypotenuse at a right angle, as shown in the diagram:

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**a** State the similarity test that the triangles $\triangle ABC$ and $\triangle ADB$ satisfy.

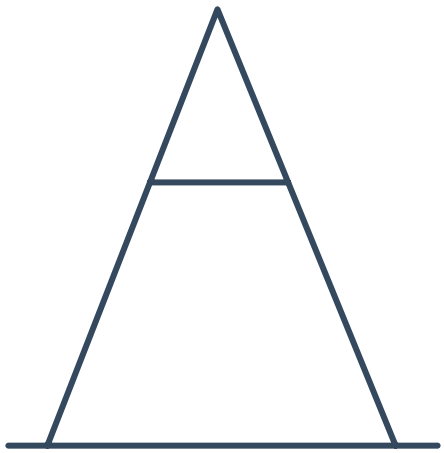
**b** How long does the brace, $BD$, need to be?

**c** Hence, find the length $x$.



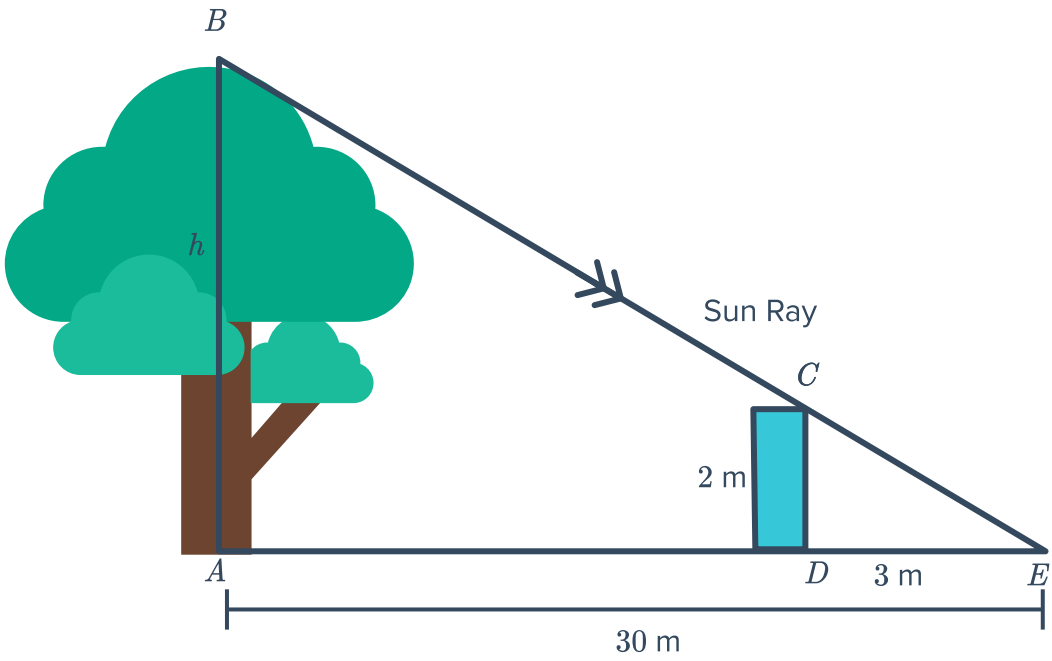

**10** A horizontal brace lies between the slanted sides of an A-shaped ladder. In this position, the top of the ladder is $220\ \text{cm}$ from the ground. The brace is $44\ \text{cm}$ in length, and sits $71\ \text{cm}$ from the top of the ladder.

How far apart are the feet of the ladder? Round your answer to the nearest whole centimetre.



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**11** A large tree casts a shadow of $30$ m. At the same time, a $2$ m high stick, standing vertically, casts a shadow of $3$ m.




**a** State the reasons why the triangles $\triangle ABE$ and $\triangle DCE$ are similar.

**b** If the height of the tree is $h$ metres, find $h$.

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